

TITLE

Does color vision improve jumping spiders' ability to find colored prey on colored backgrounds?

AUTHORS (including all who helped with project in some way, but should discuss how involved they would like to be in co-authorship process)

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INTRODUCTION

Animal eyes and visual capabilities are remarkably diverse, particularly in their ability to see and perceive color (Robledo-Ospina and Rao 2017; Land and Nilsson 2012). Color vision has evolved independently many times across taxa with many different types of eyes (Land and Nilsson 2012), suggesting that strong selection pressure exists for animals to be able to discriminate different colors (Kelber and Jacobs 2016; van der Kooi et al. 2021; XX add more from different taxa). The context in which color vision has evolved is likely due to different selection pressures like mate choice (XX) or foraging (XX). In foraging, color vision may help improve foraging capabilities (Whitfield et al. 2013; Osorio and Vorobyev 2008; Kinoshita and Stewart 2020; Troscianko et al. 2017). The ability to see certain colors is commonly associated with using specific color cues in foraging for highly specialized animals (i.e., pollinators and flowers XX, guppies XX, other examples?). On the contrary, generalist predators may be able to use color vision to detect a wide range of variously colored prey faster and more easily than other predators. Overall, color vision can improve foraging efficiency (XX), a trait that may be linked to increased reproductive output (XX). The relationship between the benefits of color vision in foraging for predators and its translation to fitness benefits is poorly understood.

Predators that primarily use visual cues in foraging may encounter many different types of prey with varying contrast to their background. Chromatic and/or achromatic coloration exhibited by potential prey that highly contrasts with the background increases prey detection (Prete 1992; Uetz et al. 2024; Donohue et al. 2020) and prey capture success for predators (de Alcantara Viana et al. 2022; Nokelainen et al. 2022). One specific benefit that animals with color vision may have is that they potentially can detect a wider range of types of contrast. Predators that can see long wavelength colors can therefore use chromatic cues to detect long wavelength colored prey on green foliage (and vice versa) during foraging, while those without color vision are limited to achromatic cues only and might not detect these contrasts at all.

Salticidae is a large family of roughly 6000 species of jumping spiders, highly visual predators, in which color vision has evolved independently at least three times (Zurek et al. 2015; Outomuro et al 2019). For example, in the genus *Habronattus*, a red spectral filter has evolved to allow spiders to see and discriminate long wavelength coloration that may be used for assessing colorful mates and making foraging decisions (Taylor and McGraw 2013; Taylor et al. 2014; Zurek et al. 2015). Within a foraging context, there exists two non-mutually exclusive alternatives as to the benefits of color vision in salticids. Previous and ongoing work has

focused on the idea that long wavelength color vision helps jumping spiders detect and avoid red aposematic prey (Vickers and Taylor 2020; Taylor et al. 2016; Powell et al. 2019; Vickers and Taylor 2018). The second benefit to color vision in foraging, and what we explored in this study, might be the ability to discriminate colorful, palatable prey against naturally-colored backgrounds (e.g., a green insect resting on an orange leaf). Understanding the fitness benefits of color vision in salticids can help us understand why color vision has independently evolved multiple times in this group.

To test the hypothesis that color vision aids in foraging by improving a spider's ability to detect and attack prey that chromatically contrast with their background, and therefore improve fitness, we compared the foraging abilities and long term reproductive output of two closely related species of jumping spiders in foraging environments with and without color cues available. The target species vary in their color vision capabilities. *Habronattus pyrrithrix* has trichromatic color vision and can see long wavelength color (Zurek et al. 2015), while the other, *Anasaitis canosus*, has dichromatic vision without the ability to see long wavelength color (XX). *H. pyrrithrix* is found primarily in grassy fields and feeds generally on many different types of insects (Taylor et al. 2019; Coco and Taylor 2017). *A. canosus* is found primarily in leaf litter and roughly 50% of their diet consists of ants (Schadegg et al. 2021; unpublished data). However, *A. canosus* is also observed feeding on other insects and spiders more generally (Schadegg et al. 2021; unpublished data). Despite the diet specialization in *A. canosus*, both species we tested here likely encounter and feed on many different potential prey that chromatically contrast with their background. We predicted that, because of their color vision, *Habronattus pyrrithrix* would have improved foraging success when color cues during foraging were available, and that improved foraging would translate into increased reproductive output. Comparatively, we predicted that having color cues available during foraging would not improve foraging success or reproductive output for *Anasaitis canosus* because they lack the color vision needed to detect color cues.

METHODS

The conceptualization, methods, and statistical analyses for this experiment were pre-registered on Open Science Framework (Lietzenmayer et al. 2023).

Collection and maintenance

We hand collected immature female and any life stage male *Habronattus pyrrithrix* in the spring of 2023 from private locations in Arizona, USA. Immature female and any life stage male *A. canosus* were collected throughout 2024 in Gainesville, Florida, USA. In the lab, females were housed in clear plastic boxes (XXX) with a top mesh hole (XXX) for ventilation and an artificial green plant (Ashland Fern Collection, Michael's Stores, Irving, TX, USA) glued to the inside of the enclosure for enrichment (Carducci and Jakob 2000). Male spider enclosures (XXX) also had a mesh hole for ventilation (XXX) and the same artificial plant. All spiders were fed pinhead crickets (*Gryllobates sigillatus*) approximately equal to their body size and enclosures misted three times per week. We monitored immature females daily for mature molts. Adult females were

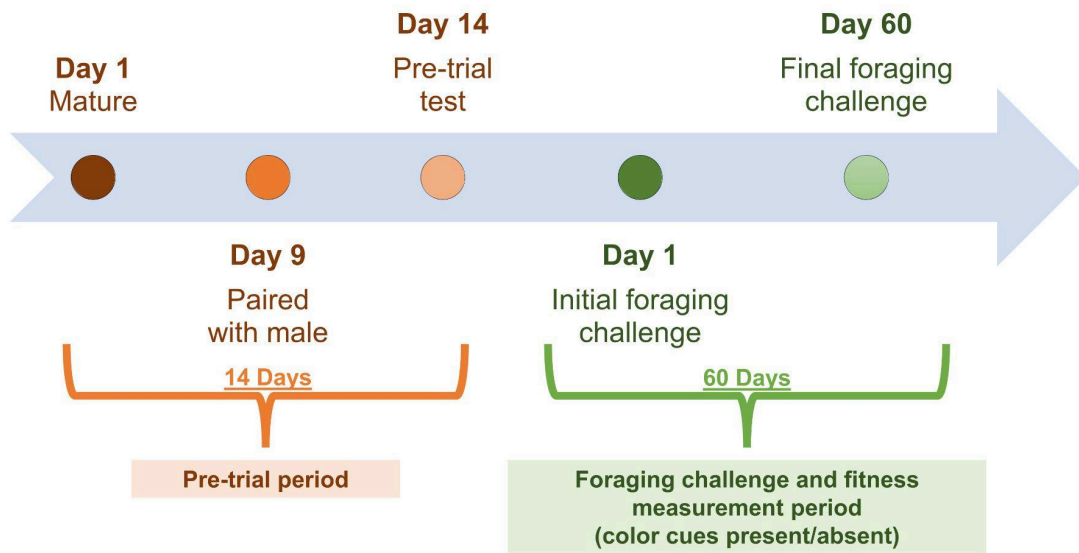
rehoused into clear plastic boxes (10 cm X 10 cm X 5 cm) with a mesh ventilation hole (2 cm diameter) the day following maturity.

Manipulation of prey color

We hand collected termites (*Reticulitermes flavipes*) from various locations in Gainesville, Florida and maintained them in the lab in large plastic buckets (XXX) with ample wood from which they were collected and cardboard for feeding. Moistened dish sponges were laid inside buckets and buckets were misted daily to ensure desiccation did not occur.

Because our understanding of brightness perception of jumping spiders is limited, we created three brightness levels (referred to as light, medium, and dark) of each color we painted the termites (red, orange, green). We matched brightness levels across the three main colors (described below). For example, the *H. pyrrithrix* visual system may perceive green as brighter than orange and red, even when the mean brightness for all three is matched using a spectrophotometer. If *H. pyrrithrix* only attacked orange and red termites, would this be because they were using chromatic cues to find orange and red termites more easily, or because they were more drawn to darker shaded prey compared to lighter shaded (i.e., green)? By manipulating all three colors to include three widely varied levels of brightness, we could assess whether spiders were using chromatic cues (and not just brightness cues) to forage.

For all color mixtures, we used gloss enamel paint (Testors, Rust-Oleum, Vernon Hills, IL, USA) and added black or white paint to manipulate brightness levels. The base green mixture consisted of 1:2 parts green to yellow paint and the base orange mixture consisted of 1:3 parts red to tangerine paint (names provided on Testors paint jars). The base red mixture was unmanipulated from the original red. Using a UV-vis spectrophotometer (USB 2000 with PX-2 pulsed xenon light source, Ocean Optics, Dunedin, FL), we measured 10 samples of each color at each brightness level to confirm that colors are statistically the same within each brightness shade (light green=light orange=light red) and different among brightness levels (for example, light green>medium green>dark green; Table 1). Overall brightness calculations were taken from within the spiders' visible range of 280-700nm (Peaslee and Wilson 1989; Li et al. 2008). Presentation of painted termites to spiders occurred after a minimum of 30 minutes to ensure paint is fully dry and any volatiles from paint dissipated.



Extinguishing learned color biases from field-collected females

Some jumping spider populations, particularly in the genus *Habronattus*, have natural aversions to aposematic colored prey (Powell et al. 2019; Taylor et al. 2014). However, jumping spiders can be trained to be averse or prefer certain colored prey by manipulating prey color and/or palatability (Taylor et al. 2016). To reduce biases in our tests from natural color aversions, we fed female spiders a diet of one green, one orange, and one red termite (all palatable) of randomly selected brightness every other day, starting on the day following female maturation. All three painted termites were placed in female home enclosures on a petri dish (XXX) lined with moist filter paper (XXX) for 24 hours. Females followed this diet for 14 days. All females attacked and consumed these palatable red and orange prey, suggesting that any natural aversion to these colors should dissipate within the pre-trial period (Taylor et al. 2016). This diet continued during the time period that females were paired with males to mate (described below). If a feeding day fell on the same day a female was paired with a male, the female was presented with the termites for one hour prior to being presented with a male (and the termites removed from the enclosure).

On day 14 post maturation, we gave females a pre-trial prey choice test of one green, one orange, and one red termite of randomly selected brightness category that followed the same set-up as described above for regular termite feedings. We recorded the color of the first termite the female attacked and consumed. If the spider did not attack within two hours, the termites were removed from the enclosure and we retested with a different randomly selected set of termites the following day. If the spider did not attack and consumed a termite in the retest, they continued on to their assigned foraging environment treatment (single color or multicolor), and were omitted from the pre-trial color bias test analyses.

Mating

Adult female *H. pyrrithrix* (n=63) were paired with a randomly selected adult male daily, 9-16 days post maturation when they are most receptive (suggested by unpublished data). Females and males were paired by introducing the male into the female enclosure for 8 hours (during daylight hours, between 8am and 6pm) and observed for mating using video recording (XXX). If mating did not occur within the allotted time period, females were randomly paired with a different male the following day until mating occurred. Males were only mated once unless no unique males were available (n=5). When pre-copulatory female sexual cannibalism occurred (n=15 observations) during the 8 hour period, the consumed male was replaced with a different male on the same day. Males that were cannibalized following successful mating (n=7) were not replaced with new males. We used the same mating protocol for adult female *A. canosus* (n=XX), and did not observe any cannibalism (THUS FAR).

Assessing whether color vision improves foraging success

After females successfully mated and completed the 14-day pre-trial feeding regime, they were randomly placed into one of two treatment groups of varying foraging scenarios (i.e., single-color vs. multicolor). Both treatments consisted of a clear plastic arena (10 cm X 10 cm X 5 cm) with a mesh hole for ventilation on the side (2 cm diameter) and a light brown fabric substrate covered with 10 grams of evenly spaced green foliage (Ashland Fern Collection, Michael's Stores, Irving, TX, USA). Each arena had a circular hole (1 cm diameter) directly in the middle of the top panel for introducing termites and the spider. In the single-color treatment, we gave female spiders a foraging environment that consisted of green colored termites on a green background (absent of chromatic cues). Spiders were presented with nine green termites, three of each level of brightness (light green, medium green, dark green; manipulation described above). In the multicolor treatment, spiders had the same green background as the single-color treatment, but the prey chromatically contrasted from the background. We also gave the spiders in the multicolor treatment nine termites, but instead included three green, three orange, and three red termites (one of each level of brightness). This allowed for both treatment groups to have three termites at each brightness level, regardless of color. In both treatments, spiders had two hours to forage, three times per week for 60 days. We video recorded (XX) the first foraging attempt on day one and the last foraging attempt on day 60 for behavioral analyses. The main behaviors we recorded to assess whether color vision improves foraging success when color cues are present included the amount of time it took for spiders to make their first attack (in minutes), the total number of termites attacked, and the total number of termites consumed. We also recorded the color and brightness of the termite first attacked and the associated color and brightness of all termites attacked within the two hour feeding period. All feedings took place within natural daylight hours between 8am and 6pm in front of large windows to allow for natural UV light. We supplemented the natural light from the windows with overhead full spectrum light bulbs (XX).

Assessing whether color vision improves fitness

To assess whether color vision has potential fitness benefits as a result of improving foraging success for *H. pyrrithrix* (and not for *A. canosus*), we used several reproductive traits as proxy measurements for overall fitness. These included the amount of time (in days) to the first egg sac production, whether females produced an egg sac at all during the 60-day period (yes or

no), the number of egg sacs produced by day 60, the total number of eggs produced by day 60, the total number of successful spiderlings emerging from all egg sacs produced within the 60 day period (including spiderlings that emerged after 60 days), and female weight on day one and day 60. We counted any leftover eggs in egg sacs after spiderlings had emerged and dispersed, then added the number of leftover eggs to the number of spiderlings that emerged to get the total number of eggs produced.

Statistical analyses

Extinguishing learned color biases from field-collected females

To confirm that we successfully eliminated any color preferences or aversions spiders brought to the lab from the field, we used a chi-square goodness of fit test to determine if spiders ate each termite color (red, orange, green) in equal proportions.

Assessing whether color vision improves foraging success

All analyses assessing improved foraging success with color cues present used data from the day one foraging trial. We used 1-tailed nonparametric Mann Whitney U tests for all analyses because our data did not meet the assumptions of a 1-tailed t-test. We tested the directional prediction that *H. pyrrithrix* would attack their first termite more quickly in the multicolor prey treatment compared to the single-color prey treatment. We also tested the directional prediction that *H. pyrrithrix* would attack more termites in the multi-color prey treatment compared with the single-color prey treatment. Lastly, we tested the directional prediction that *H. pyrrithrix* would consume more termites in the multi-color prey treatment compared with the single-color prey treatment.

Assessing whether color vision improves fitness

We used a 1-tailed t-test to test the directional prediction that females in the multicolor prey treatment would produce more eggs overall (unsuccessful and successfully spiderlings) compared to females in the single color prey treatment. For the rest of our analyses, we used a series of 1-tailed nonparametric Mann Whitney U tests because our data did not meet assumptions of a 1-tailed t-test. These analyses tested the following directional predictions that *H. pyrrithrix* females would produce their first egg sac more quickly, produce more egg sacs within the 60-day period, and produce more successful spiderlings from egg sacs laid by day 60 in the multicolor prey treatment compared to the single-color prey treatment. To determine if females were more likely to produce any egg sac in the multicolor prey treatment compared to the single color prey treatment during the 60-day period, we used a 1-tailed 2-sample Z test for proportions with eggs produced (yes or no) as the response variable.

These reproductive metrics are likely related to the time since the female mated (i.e., spiders that mated earlier during the pre-trial period would likely produce their first egg sac earlier in the 60-day period regardless of treatment). Because of this, we also ran a second set of linear models that addressed the same questions, with foraging environment (multi vs. single color) as the explanatory variable and days between mating and the beginning of the 60-day period as the covariate. The response variables included time to produce first egg sac (days), number of egg sacs produced, number of eggs produced, and number of successful spiderlings.

Exploratory analyses

To determine if spiders had any biases to different brightness levels on the termites regardless of color in the pre-trial period, we tested whether spiders attacked termites of the different brightness levels (light, medium, dark) at equal rates using a chi-square goodness of fit test that used hypothesized proportions of attacked on each brightness level that reflected the proportions of each brightness level used in the pre-trial foraging tests regardless of color.

For the day one foraging test, we determined whether *H. pyrrithrix* females in the multicolor treatment attacked red and orange termites more than green termites using a generalized linear mixed model with termite color (red, orange, green) as the predictor variable and the number of termites eaten of that color as the response variable. Because each trial has values for each of the three colors, we included trial ID as a random factor. For the females in the multicolor treatment, we also tested whether the first attack on the day one foraging test was biased towards a red or orange termite compared to a green termite using a chi-square goodness of fit test. We repeated these two analyses for the day 60 foraging tests to see if any changes had occurred during the 60-day period.

For females in both treatments, we tested whether the first attack on the day one foraging test was biased towards any brightness level (light, medium, dark) using a chi-square goodness of fit test.

We ran all analyses in RStudio using R version 4.0.3 (R Core Team, 2023). We used histograms and qqPlots of residuals to check normality of residuals and homogeneity of variance for linear models. [P-values were calculated with the Anova function in the car package \(Fox & Weisberg, 2019\)](#). We used the `glmer()` function to run generalized linear mixed models and the `lm()` function to run linear models. [This section is work in progress.](#)

All raw data are freely available on Dryad (XXX).

RESULTS [\(need to add sample sizes for analyses\)](#)

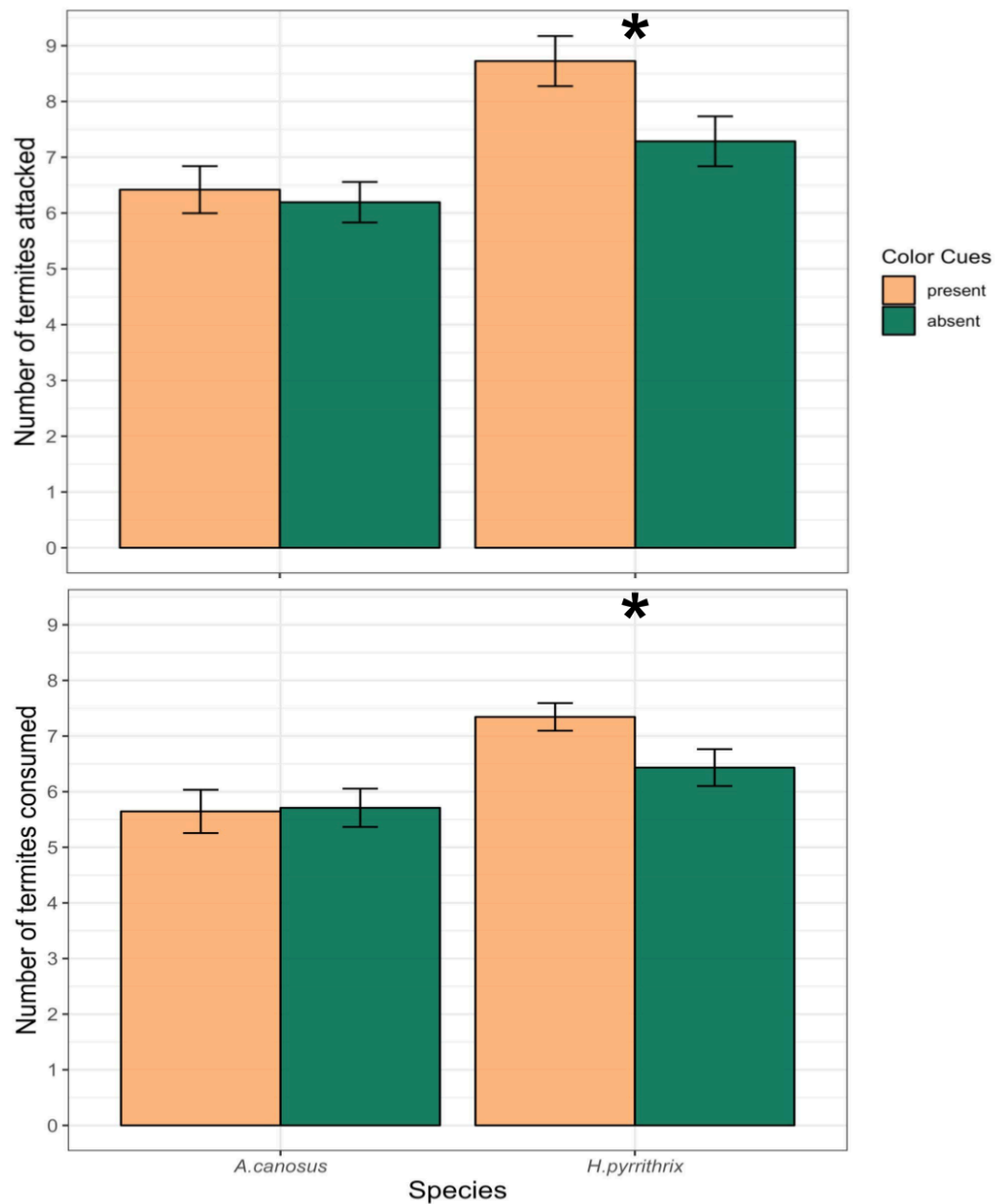
Pre-trial test to extinguish color biases

Female spiders showed no color biases for red, orange, nor green in the pre-trial test after the initial 14-day pre-trial feeding regimen to extinguish color biases from the field ($\chi^2=0.032$, $df=2$, $p=0.984$).

Assessing whether color vision improves foraging success

In the foraging test on day one of the 60-day period, females in the multicolor prey treatment attacked more termites than females in the single color prey treatment (multicolor: 8.72 ± 0.45 ; single color: 7.29 ± 0.45 ; $W=537$, $p=0.0173$). While females in the multicolor treatment consumed more termites overall than females in the single color treatment within the two hour foraging period (multicolor: 7.34 ± 0.25 termites; single color: 6.43 ± 0.33 termites; $W=616.5$,

$p=0.026$), they did not differ in the time to their first attack (multicolor: 2.17 ± 0.52 min; single color: 2.22 ± 1.16 min; $W=392.5$, $p=0.11$).



(Figure 1. For Day 1 of foraging challenge, compared species)

Assessing whether color vision improves fitness

Our prey color foraging treatments showed no significant effects for any proxies for reproductive fitness that we measured. Prey color treatment did not affect the time it took female spiders to produce their first egg sac (multicolor: 17.24 ± 2.00 days; single color: 18.73 ± 2.41 days; $W=297.5$, $p=0.305$). The proportion of females that produced at least one egg sac within the

60-day period was also comparable for spiders in the multicolor treatment (proportion=0.781) and single color treatment (proportion=0.867; $\chi^2=0.77$, $df=1$, $p=0.811$). Females also produced similar numbers of egg sacs during the 60-day period regardless of their prey color foraging environment (multicolor: 2.17 ± 0.23 ; single color: 2.37 ± 0.24 ; $W=419.5$, $p=0.683$). This was also the case when controlling for the number of days since mating occurred during the pre-trial period ($\chi^2=0.383$, $p=0.536$). The total number of eggs produced was also similar for females in both prey color treatments (multicolor= 40.37 ± 4.78 eggs; single color= 43.03 ± 4.67 eggs; $t=-0.40$, $df=58$, $p=0.654$), and when controlling for the number of days since mating occurred during the pre-trial period ([issues with GLMM version of this analysis](#)). Females in the multicolor prey treatment produced similar numbers of successful spiderlings overall (34.83 ± 4.60 spiderlings) compared to females in the single color prey treatment (35.17 ± 4.25 spiderlings; $W=461.5$, $p=0.435$). ([issues with GLMM version of this analysis](#)).

Exploratory analyses:

We found that spiders in the pre-trial foraging test attacked termites of different brightness levels (light, medium, dark) at equal rates regardless of color ($\chi^2=1.32$, $df=2$, $p=0.517$).

Females in the multicolor prey treatment did not attack red or orange termites more than green termites on day 1 of the 60-day period (mean termites attacked: red= 2.44 ± 0.276 ; orange= 2.56 ± 0.283 ; green= 2.38 ± 0.272 ; $\chi^2=0.237$, $df=2$, $p=0.888$). This linear model was singular when the random effect of trial ID was included, but the results are comparable to the simplified model that does not include trial ID as a random effect ($\chi^2=0.237$, $df=2$, $p=0.889$). Females in the multicolor prey treatment also did not show a bias toward red or orange termites on their first attack on day 1 of the 60-day period ($\chi^2=1.75$, $df=2$, $p=0.417$), nor on day 60 of the 60-day period ($\chi^2=0.929$, $df=2$, $p=0.629$). We also did not find any bias toward a specific brightness level from females in either prey color treatment on day 1 of the 60-day period ($\chi^2=0.644$, $df=2$, $p=0.725$).

[Similar results for total termites attacked and consumed between treatments for Day 60 \(multicolor spiders attacking more termites, and slight trend to multicolor consuming more termites \(\$p=0.07\$ \)\)](#)